

BEHROOZ VEDADIAN

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Senior AI Engineer and Architect

AI Systems Engineer — End-to-End AI — Deep Learning Engineer
Software Architect — Applied AI Research

SUMMARY — Senior AI engineer combining learning theory, algorithmic rigor, and craftsmanship in end-to-end AI systems, from model design to production deployment.

HIGHLIGHTS

- Proposed a Graph Signal Processing–based neural architecture (GSP-NN) in a **2015 PhD proposal**
(*first-principles model design; conceptually equivalent to later attention mechanisms — see proposal, p. 35*)
- Designed and implemented Tarjomyar (**Document Layout Analysis + Machine Translation**)
(*document layout analysis, 3-person training team lead, API and UX/UI design; primary revenue channel*)
- Contributed to national neural machine translation engine
(*system design, 3-person training lead, production serving; 80M requests/day peak*)
- Contributed to Hoda OCR system used for **2006 national census in Iran**
(*MLP training, feature engineering, CPU optimization*)
- Designed ID card reader software for City Bank of Egypt
(*fast template matching + 7-gram alphabet language model*)

KEY SKILLS

- **Programming:** C++, Python, TypeScript (systems, performance-critical, and production code)
- **Deep Learning:** Model design and training with PyTorch and TensorFlow; model deployment via ONNX
- **Serving & Interfaces:** API and service using Flask, FastAPI, Oat++, and libevent; user interfaces with ReactJS
- **Mathematics:** Linear algebra, optimization, modern statistics, and matrix calculus applied in learning systems

EDUCATION

- **PhD (coursework and research completed)** in Artificial Intelligence (Machine Translation)
at *AmirKabir University of Tech.*, GPA: 19/20
- **MSc** in Communications Engineering (signal processing) at *AmirKabir University of Tech.*, GPA: 15.01/20
- **BSc** in Electrical Engineering (signal processing) at *Sharif University of Tech.*, GPA: 15.1/20
- **Math & Physics Diploma** from *National Organization for Development of Exceptional Talents*
(selective honors program), GPA: 19.5/20

WORK EXPERIENCE

- Targoman Co. Pjs. – **Researcher / Senior AI Architect / Co-founder** *hybrid* | 2015 – 2024
- Long-term ownership of neural machine translation systems from research through production
 - Architectural design and technical decision-making across model training, serving, and product integration
 - Built and maintained training, deployment, and evaluation pipelines for translation models
- Faraadid Co. Pjs. – **Researcher / Senior AI Engineer** *hybrid* | 2017
- Systems-level work on secure model serving, including model decryption and runtime integration
 - Computation graph optimization and simplification for efficient inference
 - Applied computer vision systems for real-time vehicle tracking
- AmirKabir UoT, HLT Lab – **Researcher / Senior Software Architect** *on-site* | 2013 – 2015
- Architectural design and development of large-scale statistical machine translation systems
 - Creation of core text processing infrastructure later reused in production translation platforms
- Hoda System Co. Ltd. – **AI Engineer / Application Developer** *on-site* | 2006; 2009 – 2014
- Early-career development of OCR and machine learning systems under CPU and resource constraints
 - Hands-on work across data preprocessing, feature engineering, model training, and performance optimization
 - Production debugging and reliability work in deployed document processing systems